Therapeutic Significance of Tanshinone 13

Smriti Sudha and Mrinalini Singh

Abstract

Danshen, the dried root of *Salvia miltiorrhiza*, has been used for thousands of years as a traditional Chinese remedy, mainly for the treatment of cardiovascular and cerebrovascular diseases. Tanshinone is the major bioactive component of Danshen. It is a lipophilic compound, synthesized via two different pathways: the methylerythritol phosphate (MEP) pathway in plasmids and the mevalonate (MVA) pathway in the cytosol. Many in vitro approaches can be applied to boost its production such as metabolic engineering using elicitors and recombinant DNA technology. Tanshinone and its derivatives can be administered orally, intravenously, and intraperitoneally but have poor absorption due to low solubility and membrane permeability. This can be enhanced through structural modifications of the tanshinones. Pharmaceutical properties of tanshinones include antioxidant activity, anti-inflammatory and anti-tumor activity, immune modulation, phytoestrogen, anti-fibrosis, anti-microbial activity, neuroprotection, and bone health. These properties help tanshinone play a therapeutic role in various cardiovascular diseases including cardiac fibrosis, hypertension, myocardial infarction, and atherosclerosis. Under hypoxic conditions, tanshinone protects cardiomyocytes and microvessels from damage, helps in the recovery of ventricle dysfunction, controls hypertension, and improves heart function.

Keywords

Tanshinone · Hypoxia · Cardiovascular diseases · Hypertension

S. Sudha · M. Singh (✉)
Defence Institute of Physiology and Allied Sciences (DIPAS), Defence Research and Development Organisation (DRDO), Delhi, India
e-mail: mrinalini.dipas@gov.in

R. Tulsawani, D. Vohora (eds.), *Adaptation under Stressful Environments through Biological Adjustments and Interventions*,
https://doi.org/10.1007/978-981-99-7652-2_13

13.1 Introduction

The dried root of *Salvia miltiorrhiza*, commonly known as Danshen in China, has been widely used as traditional herbal medicine since the Eastern Han dynasty, 2000 years ago. It grows in northeastern countries like China, Japan, Korea, and the Magnolia Mountain areas and is one of more than 1000 species of the *Salvia* genus across the globe, belonging to the family Labiatae. More than 200 compounds have been identified from *Salvia miltiorrhiza* (Danshen), according to the Chinese Academy of Sciences Chemical Database (www.organchem.csdb.cn) and the Chinese herbal medicine database (Wang et al. 2015).

There are around 40 lipophilic compounds and 50 hydrophilic compounds present in Danshen (Table 13.1). Among these, a few lipophilic bioactive compounds are majorly used in pharmaceutics such as tanshinone I, tanshinone IIa, tanshinone IIb, dihydrotanshinone, tanshinlactone, and cryptotanshinone (Dong et al. 2011; Hao et al. 2016; Huang et al. 2022). Caffeic acid-derived water-soluble compound of *Salvia*, sodium tanshinone IIa sulfonate, sulfonated at C-16, is also used in pharmacology (Chen et al. 2009). Among all these, tanshinone IIa has comparatively higher therapeutic potential. Tanshinone IIA (TanIIA) abundantly occurs in the rhizomes and roots of *Salvia miltiorrhiza* (Danshen). It is widely used for various pathological conditions, such as congenital heart defects, hypertension, arrhythmia, lung disease, left ventricular remodeling, dilated coronary artery, angina pectoris, and polycystic ovary syndrome (Zhou et al. 2005; Guo et al. 2020).

Structure of Tanshinone Tanshinone, as a bioactive compound, was first isolated in 1934 by a Japanese research scholar, *Nakao*, from the root of *Salvia* (Kakisawa et al. 1968). Tanshinone is a fat-soluble compound, belonging to abietane diterpenoids. The molecular weight is 276.3 kDa. It consists of four fused rings (A, B, C, D); two of these (A and B) are tetrahydronaphthalene, ring C is paraquinone or lactone, and ring D is dihydrofuran. The overall structure of all the tanshinones is common but depending on ring saturation and different substituents present, they metabolize differently. For example, the saturated A and D rings of cryptotanshinone (CYT) are metabolized via dehydrogenation, the D ring of dihydrotanshinone I (DHTI) is metabolized through hydrolysis, and the saturated A ring of TanIIA is metabolized by hydroxylation. Alteration in the carbon substituents alters the metabolic reaction (Jiang et al. 2019; Huang et al. 2022).

13.2 Biosynthesis of Tanshinone

Tanshinone is mainly synthesized by the two universal precursors, isopentenyl pyrophosphate (IPP) (Rohmer et al. 1993), and its isomer dimethylallyl diphosphate (DMAPP), via two different pathways, as shown in Fig. 13.1 (Schwarz 1994) The methylerythritol phosphate (MEP) pathway occurs in the plastids. Since tanshinone is a diterpene, it is especially synthesized from geranylgeranyl diphosphate (Rodríguez-Concepción and Boronat 2002). Other isoterpenoids are synthesized

Table 13.1 Chemical compounds of *Salvia miltiorrhiza* (Wang et al. 2007)

S. no	Lipophilic compounds	Hydrophilic compounds
1	Tanshinone I	Denshensu (Salvianic acid A)
2	Tanshinone IIA	Protocatechuic aldehyde
3	Tanshinone IIB	Salvianic acid B
4	Cryptotanshinone	Isoferulic acid
5	Dihydrotanshinone	Methyl Rosmarinate
6	Sodium tanshinone IIA sulfonate (STS)	Rosmarinic acid
7	Methyl tanshinonate	Salvianolic acid D
8	1,2-Dihydro-tanshinquinone	Salvianolic acid F
9	Methylene-tanshinquinone	Salvianolic acid G
10	Nortanshinone	Salvianolic acid H
11	Dihydro-nortanshinone	Salvianolic acid I
12	Methyl dihydro-nortanshinonate	Salvianolic acid J
13	Hydroxytanshinone IIA	Salvianolic acid K
14	Tanshinol A and B	Salvianolic acid E
15	Przewaquinone A, B and C	Prolithospermic acid
16	3-Hydroxy tanshinone IIA	Salvinal
17	3-Hydroxymethylene tanshinquinone	Salviaflaside
18	Tanshinaldehyde	Lithospermic acid
19	Tanshindiol A, B and C	Ethyl lithospermate
20	Dihydroisotanshinone I	Dimethyl lithospermate
21	1-Ketoiso-cryptotanshinone	Isosalvianolic acid C
22	Tanshinlactone	Lithospermic acid B
23	Neo-tanshinlactone	Ethyl lithospermate B
24	Miltirone	
25	Dehydromiltirone	
26	Miltirone I	
27	Tanshinone VI	
28	Neocryptotanshinone	
29	Danshenxinkun A and Danshenxinkun B	
30	Sugiol	
31	Ferruginol	

by the mevalonate pathway (MVA pathway) in the cytosol, from 3-hydroxy-3-methylglutaryl CoA (HMG-CoA) (Miziorko 2011).

IPP isomerase (IPI) may also trigger the interconversion of IPP and DMAPP. They are then converted to geranylgeranyl diphosphate (GGPP) in a stepwise manner by geranyl diphosphate synthase (GPPS), farnesyl diphosphate synthase (FPPS), and geranylgeranyl diphosphate synthase (GGPPS). GGPP acts as the common precursor of all diterpenoids. Biosynthesis of the tanshinone skeletons is carried out by tanshinone synthases. Production of ferruginol from miltiradene was shown to be catalyzed by *CYP76AH1* through expression in yeast whole cells in vitro as well as in vivo (Cai et al. 2013). However, overexpression of key

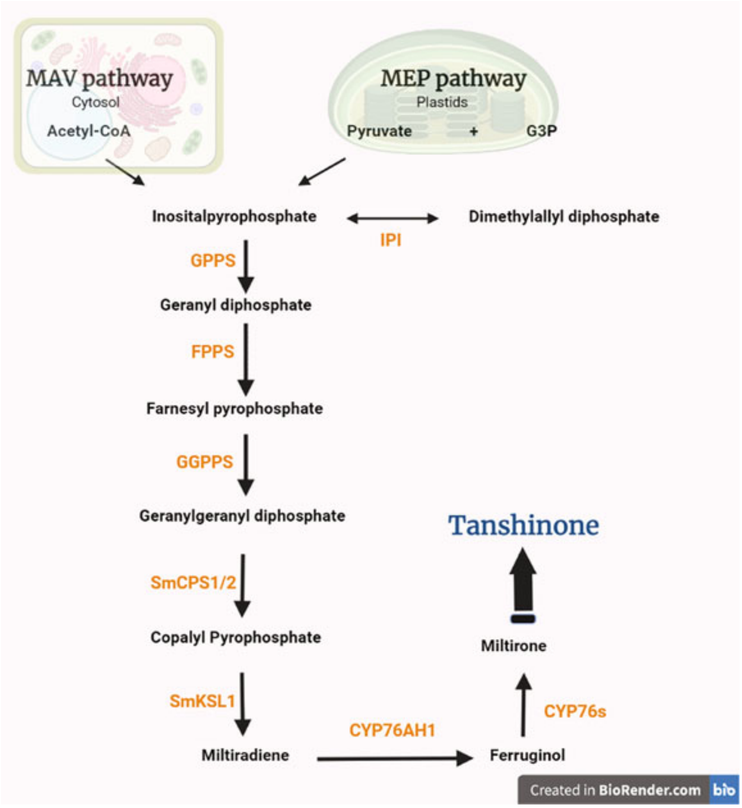

Fig. 13.1 Tanshinone biosynthesis pathway

regulatory enzymes such as 1-deoxy-D-xylulose-5-phosphate synthase (DXS) in the MEP pathway and 3-hydroxy-3-methylglutaryl-CoA reductase (HMGR) in the MAV pathway can help to enhance the tanshinone production.

13.3 Approaches to Increase Tanshinone Production

Many in vitro approaches can be used to enhance the production of tanshinone. One such approach is metabolic engineering using elicitors, which can increase tanshinone yield by approximately sixfold (Kai et al. 2012). Two types of elicitors, biotic elicitors (methylejasmonate, yeast elicitor, jasmonic acid), (Hao et al. 2015), and abiotic elicitors (Ag^{+}, Co^{2+}, and Cd^{2+}) are mainly used to elevate the genes responsible for tanshinone synthesis (Wang et al. 2016a).

Methyl jasmonate treatment can be applied in conjugation with plant hormones such as polyethyl glycol (PEG) and abscisic acid (ABA). PEG and ABA stimulate methyl jasmonate accumulation, which in turn enhances tanshinone production

(Yang et al. 2012). PEG and ABA also trigger nitric oxide (NO) formation. NO signaling plays an important part in the water stress-induced production of tanshinone. An NO donor, such as sodium nitroprusside, can also stimulate production (Du et al. 2015).

As opposed to applying a single elicitor, combining two different elicitors proves to be more successful in enhancing tanshinone production because of a synergistic effect. For example, yeast extract and hyperosmotic stress (through sorbitol addition) together increased tanshinone production remarkably more than applied separately in *Salvia miltiorrhiza* hairy root cultures (Wang and Wu 2010).

Another approach to increase the yield of tanshinone is recombinant DNA technology using *Agrobacterium rhizogenes* in hairy root cultures. There are more than 200 genes related to 32 enzymes involved in the biosynthesis of terpenoids. Overexpressing the genes identified for tanshinone can help increase the yield in vitro. The overexpression of a single *MDS* gene, which is a key regulator for tanshinone production, can result in a huge increase in tanshinone concentration. Likewise, other tanshinone-specific genes, SmGGPPS, SmDXR, SmDXS, and SmHMGR, can increase tanshinone yield by ~2.5-fold than the control.

13.4 Pharmacokinetics of Tanshinone

Tanshinone and its derivatives can be administered orally, intravenously, and intraperitoneally. It has been reported that the half-life and bio-availability of tanshinone are low, approximately 0.475–0.613 mg/g. Tan IIA has very low solubility and membrane permeability, resulting in poor absorption. The intravenous injection of tanshinone at a significant concentration (2 mg/mL) in a nanosuspension, in order to make it suitable as a parenteral solution, suggested that the Tan IIA distribution was higher in tissue (liver and lung) than in the plasma. Lipoproteins in plasma are crucial for transporting Tan IIA and other lipophilic molecules. The distribution of Tan IIA in other organs, such as the heart and brain, is comparatively lower than in the plasma (Lam et al. 2003).

However, under the pathological state of the brain, it is reported that Tan IIA can transport to the cerebral tissue due to a broken blood–brain barrier. There is a poor accumulation of Tan IIA in kidneys which may account for the meager amounts of Tan IIA and its metabolites excreted through urine. Despite being extremely lipophilic, the distribution of Tan IIA could not be seen in fat repositories. This could be due to the close association of Tan IIA with the reticuloendothelial system and the restricted blood supply to fat tissues (Hao et al. 2006).

In order to boost intestinal adsorption and solve the issue of poor solubility, structural modifications can be made to tanshinones. One such formulation is the water-soluble sodium tanshinone IIA sulfonate (STS), which shows antioxidant activity and has a wide range of pharmacological uses. Another avenue for improving the bioavailability of poorly water-soluble compounds such as tanshinones is using solid lipid nanoparticles as carriers. It has been shown to improve the circulation and residence time of Tan IIA in plasma (Table 13.2).

Table 13.2 Pharmaceutical properties of tanshinone

S. no	Therapeutic activity	Tanshinone	Mode of action	References
1.	Antioxidant activity	Tanshinone IA, Tanshinone IIA, Tanshinone IIB, Cryptotanshinone and Dihydrotanshinone I	Prevents reactive oxygen species (ROS) accumulation by inhibiting lipid peroxidation and Ca^{2+} influx in myocardial cells, and by increasing levels of superoxide dismutase (SOD) and growth stimulating hormone (GSH) Inhibits mitochondrial autophagy, and decreases malondialdehyde (MDA) and inducible nitric oxide synthase level. Increases level of anti-oxidant enzymes	Guo et al. (2018), Kubli et al. (2013), Li et al. (2006), Wang et al. (2020)
2.	Anti-inflammatory	Tanshinone IIA and Cryptotanshinone	Inhibits expression of pro-inflammatory cytokine gene, TLR2/Nf-κB signaling. Reduces expression of adhesion molecules and infiltration of neutrophils	Chen et al. (2001), Yang et al. (2016)
3.	Anti-tumor	Tanshinone IIA, Tanshinone I and Cryptotanshinone	Inhibits cancer cell growth and proliferation, promoting apoptosis by regulating p53 gene and cell cycle arrest. Inhibits E6 protein, in case of the Human Papilloma Virus (HPV) Controls acute leukemia through the tumor suppressor gene and AKT/mTOR pathways	Lu et al. (2013), Xu et al. (2018), Yintao (2010)
4.	Immune modulation	Tanshinone IIA	Induces dendritic cell-mediated innate immunity response, upregulates regulatory	Gong et al. (2020), Qin et al. (2010)

(continued)

Table 13.2 (continued)

S. no	Therapeutic activity	Tanshinone	Mode of action	References
			cytokines such as TGF-β1, promotes anti-inflammatory cytokines production and reduces LPS-induced TLR4/NF-κB signaling cascade. Induces T helper cell polarization into T reg cells	
5.	Phyto-estrogen	Tanshinone I, Tanshinone IIA, Cryptotanshinone and Dihydrotanshinone	Activates and modulates G-protein-coupled estrogen receptor to regulate transcription of collagen and elastin genes. Can also activate PXR, PPARγ, and AR	Mao et al. (2014)
6.	Anti-fibrosis	Tanshinone IIA	Decreases myofibroblast proliferation by activating Nrf2/glutathione signaling. Blocks ROS-mediated PKCδ/smad3 pathway and miRNA-mediated TGF-β1/smad3 pathway	Feng et al. (2019), Meng et al. (2016), Shu et al. (2016)
7.	Neuroprotectant	Tanshinone IIA, Tanshinone I and Cryptotanshinone	Improves learning and memory deficits by shielding neuronal loss through inhibition of ERK and GSK-3β pathway. Downregulates p53 protein and maintains membrane potential to inhibit apoptosis of neuronal cells. Protects neurons from beta-amyloid toxicity.	Bi et al. (2008), Chen et al. (2018)
8.	Anti-microbial	Dihydrotanshinone and Cryptotanshinone	Anti-bacterial activity against a wide range of Gram-positive bacteria	Honda et al. (1988)

(continued)

Table 13.2 (continued)

S. no	Therapeutic activity	Tanshinone	Mode of action	References
9.	Bone health	Tanshinone IIa	Regulates RANKL-MCSF signaling and inhibits excessive bone resorption. Controls osteoporosis	Guo et al. (2014), Lee et al. (2005), Wang et al. (2019)

13.5 Therapeutic Role of Tanshinone in Cardiovascular Diseases

13.5.1 Cardiac Fibrosis (Hinderer and Schenke-Layland 2019)

Cardiac fibrosis causes various heart diseases such as arrhythmias, hypertensive heart diseases, hypertrophic cardiomyopathy, and coronary heart diseases. It involves fibrotic scar development due to excessive and abnormal extracellular matrix (ECM) formation, especially collagen type I, which results in endomyocardial fibrosis and impaired normal tissue function (Murtha et al. 2017). Mast cell accumulation and degranulation also promote excess collagen and fibroblast synthesis (Kong et al. 2014). TanIIA inhibits NF-κB mediated signaling cascade to reduce inflammatory cytokine production (Wu et al. 2018) and upregulates matrix metalloproteinases to promote collagen metabolism (Mao et al. 2014). As protective activity against cardiac fibrosis, TanIIA further inhibits β-tubulin expression (Zhang et al. 2018b) and reduces oxidative stress causing fibroblast formation by activating Nrf2, which inhibits TGF-β generation (Bi et al. 2021; Zhou et al. 2012). Cardiac fibrosis also results in cell death due to inflammatory response, by increased TGF-β (Desmoulière et al. 1993). TanIIA has shown a significant anti-apoptotic response, by regulating caspase protein and mitochondrial membrane potential, which results in regulated cytochrome *c* secretion (Zhang et al. 2014).

13.5.2 Hypertension

Globally, one in every four adults suffers from systemic arterial hypertension, and the percentage of individuals with non-optimum blood pressure (BP) is increasing every year (Forouzanfar et al. 2017). The average range of BP is around 115/75 mmHg, which is a ratio of systolic and diastolic pressures. However, there could be an increase in normal BP due to several influences such as environment, lifestyle, excessive sodium consumption, as well as genetics (Poulter et al. 2015). Many internal factors are involved in the human body that maintain physiological blood pressure, such as the renin–angiotensin–aldosterone system, the sympathetic nervous system, as well as the immune system. Interruption of these parameters can lead to cardiovascular disease. It is reported in many studies that the onset condition of high blood pressure can be controlled by tanshinone IIA. It is observed that tanshinone inhibits angiotensin-converting enzymes to lower the blood pressure.

Sodium plays a very crucial role in regulating blood pressure by maintaining blood volume. Imbalance in sodium concentrations will lead to the production of nitric oxide to vasodilate blood vessels. However, some individuals show salt-sensitivity with endothelial dysfunction, due to which there is an overproduction of TGF-β through the voltage- and calcium-activated potassium channels (Feng et al. 2017) and MAPK pathway, leading to fibrosis and oxidative stress (Hovater and Sanders 2012). Vasoconstriction is very prominent under hypoxia since the voltage-gated potassium channel is closed, causing membrane depolarization, and thus increasing Ca^{2+} ion concentration. Tan IIA induces the PI3K/Akt-eNOS signaling pathway for vasodilation and aortic relaxation. It is also found that tanshinone opens ATP-sensitive K^{+} channels to decrease calcium ion levels, leading to vasodilation (Sonkusare et al. 2006; Chan et al. 2011). Moreover, disruption in several genes and their allelic variants can also be involved in hypertension and cardiovascular disease. To improve ventricular remodeling, Tan IIA can downregulate TGF-β1 through miR-205-3p and thus maintain physiological blood pressure (Qiao et al. 2021).

13.5.3 Myocardial Infarction

Myocardial infarction has affected more than 15 million people. It occurs due to excessive reactive oxygen species (ROS) production and inflammation; leading to cardiomyocyte death. The antioxidant and anti-inflammatory properties of TanIIA help to control myocardial injury (Lu et al. 2022). TanIIA inhibits inflammation generated during ischemia by reducing the expression of inflammatory cytokines such as MCP-1 and TGF-β1, as well as infiltration of monocytes, and macrophages to the infarcted area, to improve heart function. TanIIA has been used for years to treat myocardial diseases, as it can increase coronary flow by dilating coronary arteries, which prevents ischemia. TanIIA also shows an anti-inflammatory effect by preventing NF-κB-mediated TNF-α production (Ren et al. 2010). It has been reported that TanIIA helps recovery from MI-induced heart injury by decreasing infarct size and collagen deposition.

13.5.4 Atherosclerosis

Atherosclerosis is a multi-arterial inflammatory disease, resulting mainly from hypertension and hypercholesterolemia. It is caused by the hardening of arteries due to excessive deposition of plaque. The increased plasma lipids promote the accumulation of low-density lipids C (LDL-C) on the walls of the arteries and escalate the expression of adhesion molecules. This leads to adherence of monocytes to the sub-endothelial spaces and conversion into foamy macrophages. They express

CD68, which works as a chemoattractant for the accumulation of intracellular cholesterol (Sakakura et al. 2013).

Anti-oxidant activity of TanIIA prevents ROS-mediated LDL oxidation (Hartley et al. 2019) and the migration of monocytes into the arterial endothelium (Tabas and Bornfeldt 2016). It also increases GPx and SOD levels to avoid foamy macrophage accumulation (Tang et al. 2007) and prevents the accumulation of platelets and cholesterols (Ahmadsei et al. 2015) through anti-inflammatory activity.

Overexpression of Nf-κB causes an inflammatory response; thus, blocking the activation of Nf-κB could help cure inflammation-mediated atherosclerosis. Research findings confirmed that TanIIA has the potential to regulate Nf-κb activation under diseased conditions by attenuating the expression of adhesion molecules such as VCAM and ICAM (Chang et al. 2014).

TanIIA also possesses immunomodulatory effects. It controls the expression of pro-inflammatory cytokines, DC maturation, and T-cell maturation (Li et al. 2014).

Calcium influx results in the mineralization of plaque, leading to the narrowing of the arteries. Under stress or hypoxic conditions, calcium influx is inhibited by TanIIA, which helps to improve energy production in cardiomyocytes and maintain cardiac functioning (Satoshi et al. 1990).

13.6 Effect of Tanshinone on Cardiovascular Diseases Under Hypoxia

High altitude is characterized by a decrease in the partial pressure of oxygen, resulting in several pathological conditions (Avellanas Chavala 2018), such as acute mountain sickness (AMS), high-altitude pulmonary edema (HAPE), high-altitude cerebral edema (HACE), and high-altitude pulmonary hypertension (HAPH) (Simonneau et al. 2009; Bärtsch and Swenson 2013). HAPH is caused by pulmonary vasoconstriction under chronic exposure to hypobaric hypoxia (Hakim et al. 1983). The mean pulmonary arterial pressure is less than 25 mmHg; however, under HAPH, it exceeds 30 mmHg (León-Velarde et al. 2005).

Vascular resistance is increased due to long-term vasoconstriction under hypobaric hypoxia (Siques et al. 2021). Vascular remodeling is triggered, which results in right ventricle hypertrophy. Vasoconstrictors, such as reactive oxygen species (ROS) and endothelin-1, also contribute to vascular remodeling. The endogenous antioxidant system becomes inefficient at removing excess ROS due to atmospheric stress at high altitudes. The enhanced ROS results in oxidative stress on the cell, causing mitochondrial dysfunction and cell death (Farías et al. 2016). In cardiomyocytes and other cells, NADPH is responsible for the production of ROS. Under hypoxia, the expression of the NADPH complex (Nox4) is increased, which induces TGF-β level causing pulmonary hypertension (Sturrock et al. 2006; Zhang et al. 2018a, b; Siques et al. 2021). Vascular remodeling is regulated by MAPK signaling and the activator protein-1 (AP-1) transcription factor which controls proliferation, inflammation, and apoptosis. Thus, AP-1 subunits could be a good therapeutic target (Yadav et al. 2017; Singh et al. 2018). Tanshinone IIA has been shown to modulate and promote cell survival in human lung epithelial cells under hypoxia via the AP-1-Nrf2 transcription factor (Yadav et al. 2020).

During hypoxia, cardiomyocytes undergo apoptosis and narcosis due to the activation of specific pathways. Excessive apoptosis results in cardiovascular disorders such as heart failure and myocardial infarction (Chen et al. 2014). TanIIA protects cardiomyocytes, and microvessels from damage from excessive superoxide production, and intracellular calcium influx, under hypoxia. Left ventricle dysfunction, resulting due to chronic intermittent hypoxia, is also recovered by TanIIA (Wang et al. 2017).

Activation of hypoxia-induced factor (HIF) is required to maintain normal physiology under hypoxia (Forsythe et al. 1996). In the case of cardiovascular diseases, TanIIA improves heart function by upregulating HIF and further enhancing angiogenesis (Xu et al. 2009). However, in cancer conditions, TanIIA exhibits anti-angiogenesis by decreasing the HIF-1α level and inhibiting growth factors like VEGH and β-FGF (Sui et al. 2017).

Classical transient receptor potential canonical (TRPC) channels are closely associated with cardiac function and play a role in cardiac hypertrophy, myocardial infarction, arrhythmias, and fibrosis. Under chronic hypoxia, the TanIIA derivative STS reduces Ca^{2+} influx and prevents the expression of TRPC1 and TRPC6 (Wang et al. 2013). STS also inhibits the BMPR-induced smad pathway to control unnecessary cell apoptosis. TanIIA inhibits TGF-β induced smad3 pathway to control hypertension under hypoxia. In vivo studies have reported that TanIIA can induce apoptosis to reverse the pulmonary artery remodeling by suppressing the JAK2/STAT3 pathway and enhancing the connexin 43 level (Chen et al. 2014) under hypoxia.

Various classes of non-coding RNA, such as small RNAs and long non-coding RNAs, could be potential targets for cardiovascular therapeutics as many non-coding RNAs are involved in cardiovascular diseases. miR-1 is an important non-coding RNA (miRNA) that regulates a number of genes in myocytes, including reduced expression of the apoptotic genes and Akt phosphorylation. Misregulation of miR-1 can cause myoblast apoptosis, arrhythmogenesis, cardiac hypertrophy, etc. A study found that TanIIA can suppress miR-1 expression by inhibiting p38 MAPK, which improves heart function during ischemia and hypoxic injury (Zhang et al. 2010). miR-133 regulates apoptosis-related proteins such as caspase9 and reduced apoptosis rate can be observed in cardiac cells by upregulating miR-133 mediated MAPK ERK1/2 expression. This can be activated by both hypoxia and TanIIA.

Also, hypoxia followed by re-oxygenation results in the upregulation of pro-apoptotic pathways. However, pre-treatment with TanIIA increases cell viability, decreases the expression of pro-apoptotic genes, and protects against hypoxia-induced damage (Cui et al. 2016).

References

Ahmadsei M, Lievens D, Weber C, von Hundelshausen P, Gerdes N (2015) Immune-mediated and lipid-mediated platelet function in atherosclerosis. Curr Opin Lipidol 26(5):438–448. https://doi.org/10.1097/MOL.0000000000000212, PMID 26270811

Avellanas Chavala ML (2018) Un viaje entre la hypoxia de la gran altitud y la hypoxia del enfermo crítico: ¿quépuede enseñarnos en la compresión y manejo de las enfermedades críticas? Med Intensiva 42(6):380–390. https://doi.org/10.1016/j.medin.2017.08.006, PMID 28919307

Bärtsch P, Swenson ER (2013) Clinical practice: acute high-altitude illnesses. N Engl J Med 368(24):2294–2302. https://doi.org/10.1056/NEJMcp1214870, PMID 23758234

Bi HC, Zuo Z, Chen X, Xu CS, Wen YY, Sun HY et al (2008) Preclinical factors affecting the pharmacokinetic behaviour of tanshinone IIA, an investigational new drug isolated from Salvia miltiorrhiza for the treatment of ischaemic heart diseases. Xenobiotica 38(2):185–222. https://doi.org/10.1080/00498250701767675, PMID 18197559

Bi Z, Wang Y, Zhang W (2021) A comprehensive review of tanshinone IIA and its derivatives in fibrosis treatment. Biomed Pharmacother 137:111404. https://doi.org/10.1016/j.biopha.2021.111404, PMID 33761617

Cai Y, Guo J, Zhou YJ, Zhu ZW, Wu WY, Huang LQ et al (2013) [Construction of Saccharomyces cerevisiae whole-cell biocatalyst system for conversion miltiradiene]. Yao Xue Xue Bao 48(10): 1618–1623, PMID 24417091

Chan P, Liu IM, Li YX, Yu WJ, Cheng JT (2011) Antihypertension induced by tanshinone IIA isolated from the roots of Salvia miltiorrhiza. Evid Based Complement Alternat Med 2011: 392627. https://doi.org/10.1093/ecam/nep056, PMID 19542183

Chang CC, Chu CF, Wang CN, Wu HT, Bi KW, Pang JH et al (2014) The anti-atherosclerotic effect of tanshinone IIA is associated with the inhibition of TNF-α-induced VCAM-1, ICAM-1 and CX3CL1 expression. Phytomedicine 21(3):207–216. https://doi.org/10.1016/j.phymed.2013.09.012, PMID 24157079

Chen YH, Lin SJ, Ku HH, Shiao MS, Lin FY, Chen JW et al (2001) Salvianolic acid B attenuates VCAM-1 and ICAM-1 expression in TNF-alpha-treated human aortic endothelial cells. J Cell Biochem 82(3):512–521. https://doi.org/10.1002/jcb.1176, PMID 11500927

Chen Y, Tu JH, He YJ, Zhang W, Wang G, Tan ZR et al (2009) Effect of sodium tanshinone II A sulfonate on the activity of CYP1A2 in healthy volunteers. Xenobiotica 39(7):508–513. https://doi.org/10.1080/00498250902951763, PMID 19534587

Chen M, Liu Y, Yi D, Wei L, Li Y, Zhang L (2014) Tanshinone IIA promotes pulmonary artery smooth muscle cell apoptosis in Vitroby inhibiting the JAK2/STAT3 signaling pathway. Cell Physiol Biochem 33(4):1130–1138. https://doi.org/10.1159/000358682, PMID 24733040

Chen J, Bi Y, Chen L, Zhang Q, Xu L (2018) Tanshinone IIA exerts neuroprotective effects on hippocampus-dependent cognitive impairments in diabetic rats by attenuating ER stress-induced apoptosis. Biomed Pharmacother 104:530–536. https://doi.org/10.1016/j.biopha.2018.05.040, PMID 29800917

Cui ZT, Liu JP, Wei WL (2016) The effects of tanshinone IIA on hypoxia/reoxygenation-induced myocardial microvascular endothelial cell apoptosis in rats via the JAK2/STAT3 signaling pathway. Biomed Pharmacother 83:1116–1126. https://doi.org/10.1016/j.biopha.2016.07.054, PMID 27551758

Desmoulière A, Geinoz A, Gabbiani F, Gabbiani G (1993) Transforming growth factor-beta 1 induces alpha-smooth muscle actin expression in granulation tissue myofibroblasts and in quiescent and growing cultured fibroblasts. J Cell Biol 122(1):103–111. https://doi.org/10.1083/jcb.122.1.103, PMID 8314838

Dong Y, Morris-Natschke SL, Lee KH (2011) Biosynthesis, total syntheses, and antitumor activity of tanshinones and their analogs as potential therapeutic agents. Nat Prod Rep 28(3):529–542. https://doi.org/10.1039/c0np00035c, PMID 21225077

Du X, Zhang C, Guo W, Jin W, Liang Z, Yan X et al (2015) Nitric oxide plays a central role in water stress-induced tanshinone production in Salvia miltiorrhiza hairy roots. Molecules 20(5): 7574–7585. https://doi.org/10.3390/molecules20057574, PMID 25919278

Farías JG, Herrera EA, Carrasco-Pozo C, Sotomayor-Zárate R, Cruz G, Morales P et al (2016) Pharmacological models and approaches for pathophysiological conditions associated with hypoxia and oxidative stress. Pharmacol Ther 158:1–23. https://doi.org/10.1016/j.pharmthera.2015.11.006, PMID 26617218

Feng W, Dell'Italia LJ, Sanders PW (2017) Novel paradigms of salt and hypertension. J Am Soc Nephrol 28(5):1362–1369. https://doi.org/10.1681/ASN.2016080927, PMID 28220030

Feng F, Cheng P, Zhang H, Li N, Qi Y, Wang H, Wang Y, Wang W (2019) The protective role of tanshinone IIA in silicosis rat model via TGF-β1/Smad signaling suppression, NOX4 inhibition and Nrf2/ARE signaling activation. Drug Des Devel Ther 13:4275–4290. https://doi.org/10.2147/DDDT.S230572

Forouzanfar MH, Liu P, Roth GA, Ng M, Biryukov S, Marczak L et al (2017) Global burden of hypertension and systolic blood pressure of at least 110 to 115 mmHg, 1990-2015. JAMA 317(2):165–182. https://doi.org/10.1001/jama.2016.19043, PMID 28097354

Forsythe JA, Jiang BH, Iyer NV, Agani F, Leung SW, Koos RD et al (1996) Activation of vascular endothelial growth factor gene transcription by hypoxia-inducible factor 1. Mol Cell Biol 16(9): 4604–4613. https://doi.org/10.1128/MCB.16.9.4604, PMID 8756616

Gong Y, Liu YC, Ding XL, Fu Y, Cui LJ, Yan YP (2020) Tanshinone IIA ameliorates CNS autoimmunity by promoting the differentiation of regulatory T cells. Neurotherapeutics 17(2): 690–703. https://doi.org/10.1007/s13311-019-00789-2, PMID 31845175

Guo Y, Li Y, Xue L, Severino RP, Gao S, Niu J et al (2014) Salvia miltiorrhiza: an ancient Chinese herbal medicine as a source for anti-osteoporotic drugs. J Ethnopharmacol 155(3):1401–1416. https://doi.org/10.1016/j.jep.2014.07.058, PMID 25109459

Guo Z, Yan M, Chen L, Fang P, Li Z, Wan Z et al (2018) Nrf2dependent antioxidant response mediated the protective effect of tanshinone IIA on doxorubicin induced cardiotoxicity. Exp Ther Med 16(4):3333–3344. https://doi.org/10.3892/etm.2018.6614, PMID 30233680

Guo R, Li L, Su J, Li S, Duncan SE, Liu Z et al (2020) Pharmacological activity and mechanism of Tanshinone IIA in related diseases. Drug Des Dev Ther 14:4735

Hakim TS, Michel RP, Minami H, Chang HK (1983) Site of pulmonary hypoxic vasoconstriction studied with arterial and venous occlusion. J Appl Physiol Respir Environ Exerc Physiol 54(5): 1298–1302. https://doi.org/10.1152/jappl.1983.54.5.1298, PMID 6863090

Hao H, Wang G, Cui N, Li J, Xie L, Ding Z (2006) Pharmacokinetics, absorption and tissue distribution of tanshinone IIA solid dispersion. Planta Med 72(14):1311–1317. https://doi.org/10.1055/s-2006-951698, PMID 17024606

Hao X, Shi M, Cui L, Xu C, Zhang Y, Kai G (2015) Effects of methyl jasmonate and salicylic acid on tanshinone production and biosynthetic gene expression in transgenic Salvia miltiorrhiza hairy roots. Biotechnol Appl Biochem 62(1):24–31. https://doi.org/10.1002/bab.1236, PMID 24779358

Hao W, Chen L, Wu LF, Yang F, Niu JX, Kaye AD et al (2016) Tanshinone IIA exerts an antinociceptive effect in rats with cancer-induced bone pain. Pain Phys 19(7):456–476, PMID 27676663

Hartley A, Haskard D, Khamis R (2019) Oxidized LDL and anti-oxidized LDL antibodies in atherosclerosis—novel insights and future directions in diagnosis and therapy. Trends Cardiovasc Med 29(1):22–26. https://doi.org/10.1016/j.tcm.2018.05.010, PMID 29934015

Hinderer S, Schenke-Layland K (2019) Cardiac fibrosis—a short review of causes and therapeutic strategies. Adv Drug Deliv Rev 146:77–82. https://doi.org/10.1016/j.addr.2019.05.011, PMID 31158407

Honda G, Koezuka Y, Tabata M (1988) Isolation of an antidermatophytic substance from the root of Salvia miltiorrhiza. Chem Pharm Bull (Tokyo) 36(1):408–411. https://doi.org/10.1248/cpb.36.408, PMID 3378301

Hovater MB, Sanders PW (2012) Effect of dietary salt on regulation of TGF-β in the kidney. Semin Nephrol 32(3):269–276. https://doi.org/10.1016/j.semnephrol.2012.04.006, PMID 22835458

Huang X, Jin L, Deng H, Wu D, Shen QK, Quan ZS et al (2022) Research and development of natural product tanshinone I: pharmacology, total synthesis, and structure modifications. Front Pharmacol 13:920411. https://doi.org/10.3389/fphar.2022.920411, PMID 35903340

Jiang Z, Gao W, Huang L (2019) Tanshinones, critical pharmacological components in Salvia miltiorrhiza. Front Pharmacol 10:202. https://doi.org/10.3389/fphar.2019.00202, PMID 30923500

Kai G, Liao P, Xu H, Wang J, Zhou C, Zhou W et al (2012) Molecular mechanism of elicitor-induced tanshinone accumulation in Salvia miltiorrhiza hairy root cultures. Acta Physiol Plant 34(4):1421–1433. https://doi.org/10.1007/s11738-012-0940-z

Kakisawa H, Hayashi T, Okazaki I, Ohashi M (1968) Isolation and structures of new tanshinones. Tetrahedron Lett 9(28):3231–3234. https://doi.org/10.1016/S0040-4039(00)89532-5

Kong P, Christia P, Frangogiannis NG (2014) The pathogenesis of cardiac fibrosis. Cell Mol Life Sci 71(4):549–574. https://doi.org/10.1007/s00018-013-1349-6, PMID 23649149

Kubli DA, Zhang X, Lee Y, Hanna RA, Quinsay MN, Nguyen CK et al (2013) Parkin protein deficiency exacerbates cardiac injury and reduces survival following myocardial infarction. J Biol Chem 288(2):915–926. https://doi.org/10.1074/jbc.M112.411363, PMID 23152496

Lam BYH, Lo ACY, Sun X, Luo HW, Chung SK, Sucher NJ (2003) Neuroprotective effects of tanshinones in transient focal cerebral ischemia in mice. Phytomedicine 10(4):286–291. https://doi.org/10.1078/094471103322004776, PMID 12809358

Lee SY, Choi DY, Woo ER (2005) Inhibition of osteoclast differentiation by tanshinones from the root of Salvia miltiorrhiza Bunge. Arch Pharm Res 28(8):909–913. https://doi.org/10.1007/BF02973876, PMID 16178416

León-Velarde F, Maggiorini M, Reeves JT, Aldashev A, Asmus I, Bernardi L et al (2005) Consensus statement on chronic and subacute high altitude diseases. High Alt Med Biol 6(2): 147–157. https://doi.org/10.1089/ham.2005.6.147, PMID 16060849

Li P, Wang GJ, Li J, Hao HP, Zheng CN (2006) Identification of tanshinone IIA metabolites in rat liver microsomes by liquid chromatography–tandem mass spectrometry. J Chromatogr A 1104(1–2):366–369. https://doi.org/10.1016/j.chroma.2005.12.047, PMID 16405978

Li HZ, Lu YH, Huang GS, Chen Q, Fu Q, Li ZL (2014) Tanshinone II A inhibits dendritic cell-mediated adaptive immunity: potential role in anti-atherosclerotic activity. Chin J Integr Med 20(10):764–769. https://doi.org/10.1007/s11655-012-1213-9, PMID 23001464

Lu L, Li C, Li D, Wang Y, Zhou C, Shao W et al (2013) Cryptotanshinone inhibits human glioma cell proliferation by suppressing STAT3 signaling. Mol Cell Biochem 381(1–2):273–282. https://doi.org/10.1007/s11010-013-1711-x, PMID 23740516

Lu TC, Wu YH, Chen WY, Hung YC (2022) Targeting oxidative stress and endothelial dysfunction using tanshinone IIA for the treatment of tissue inflammation and fibrosis. Oxidative Med Cell Longev 2022:2811789. https://doi.org/10.1155/2022/2811789, PMID 35432718

Mao S, Wang Y, Zhang M, Hinek A (2014) Phytoestrogen, tanshinone IIA diminishes collagen deposition and stimulates new elastogenesis in cultures of human cardiac fibroblasts. Exp Cell Res 323(1):189–197. https://doi.org/10.1016/j.yexcr.2014.02.001, PMID 24525372

Meng XM, Nikolic-Paterson DJ, Lan HY (2016) TGF-β: the master regulator of fibrosis. Nat Rev Nephrol 12(6):325–338. https://doi.org/10.1038/nrneph.2016.48, PMID 27108839

Miziorko HM (2011) Enzymes of the mevalonate pathway of isoprenoid biosynthesis. Arch Biochem Biophys 505(2):131–143. https://doi.org/10.1016/j.abb.2010.09.028, PMID 20932952

Murtha LA, Schuliga MJ, Mabotuwana NS, Hardy SA, Waters DW, Burgess JK et al (2017) The processes and mechanisms of cardiac and pulmonary fibrosis. Front Physiol 8:777. https://doi.org/10.3389/fphys.2017.00777, PMID 29075197

Poulter NR, Prabhakaran D, Caulfield M (2015) Hypertension. Lancet 386(9995):801–812. https://doi.org/10.1016/S0140-6736(14)61468-9

Qiao P, Xu J, Liu X, Li X (2021) Tanshinone IIA improves ventricular remodeling following cardiac infarction by regulating miR-205-3p. Dis Markers 2021:8740831. https://doi.org/10.1155/2021/8740831, PMID 34880957

Qin XY, Li T, Yan L, Liu QS, Tian Y (2010) Tanshinone IIA protects against immune-mediated liver injury through activation of T-cell subsets and regulation of cytokines. Immunopharmacol Immunotoxicol 32(1):51–55. https://doi.org/10.1080/08923970903120997, PMID 19653860

Ren ZH, Tong YH, Xu W, Ma J, Chen Y (2010) Tanshinone II A attenuates inflammatory responses of rats with myocardial infarction by reducing MCP-1 expression. Phytomedicine 17(3–4): 212–218. https://doi.org/10.1016/j.phymed.2009.08.010, PMID 19800776

Rodríguez-Concepción M, Boronat A (2002) Elucidation of the methylerythritol phosphate pathway for isoprenoid biosynthesis in bacteria and plastids. A metabolic milestone achieved through genomics. Plant Physiol 130(3):1079–1089. https://doi.org/10.1104/pp.007138, PMID 12427975

Rohmer M, Knani M, Simonin P, Sutter B, Sahm H (1993) Isoprenoid biosynthesis in bacteria: a novel pathway for the early steps leading to isopentenyl diphosphate. Biochem J 295(2): 517–524. https://doi.org/10.1042/bj2950517, PMID 8240251

Sakakura K, Nakano M, Otsuka F, Ladich E, Kolodgie FD, Virmani R (2013) Pathophysiology of atherosclerosis plaque progression. Heart Lung Circ 22(6):399–411. https://doi.org/10.1016/j.hlc.2013.03.001, PMID 23541627

Satoshi T, Kouichi T, Katsumi H, Kazuko K, Mika O, Akira Y et al (1990) Beneficial effect of Tan-Shen, an extract from the root of salvia, on post-hypoxic recovery of cardiac contractile force. Biochem Pharmacol 40(5):1137–1143. https://doi.org/10.1016/0006-2952(90)90504-E

Schwarz MKV (1994) Terpen-Biosynthese in Ginkgo biloba eine überraschende Geschichte

Shu M, Hu XR, Hung ZA, Huang DD, Zhang S (2016) Effects of tanshinone IIA on fibrosis in a rat model of cirrhosis through heme oxygenase-1, inflammation, oxidative stress and apoptosis. Mol Med Rep 13(4):3036–3042. https://doi.org/10.3892/mmr.2016.4886, PMID 26936326

Simonneau G, Robbins IM, Beghetti M, Channick RN, Delcroix M, Denton CP et al (2009) Updated clinical classification of pulmonary hypertension. J Am Coll Cardiol 54(1 Suppl): S43–S54. https://doi.org/10.1016/j.jacc.2009.04.012, PMID 19555858

Singh M, Yadav S, Kumar M, Saxena S, Saraswat D, Bansal A et al (2018) The MAPK-activator protein-1 signaling regulates changes in lung tissue of rat exposed to hypobaric hypoxia. J Cell Physiol 233(10):6851–6865. https://doi.org/10.1002/jcp.26556, PMID 29665093

Siques P, Pena E, Brito J, el Alam S (2021) Oxidative stress, kinase activation, and inflammatory pathways involved in effects on smooth muscle cells during pulmonary artery hypertension under hypobaric hypoxia exposure. Front Physiol 12:690341. https://doi.org/10.3389/fphys.2021.690341, PMID 34434114

Sonkusare S, Palade PT, Marsh JD, Telemaque S, Pesic A, Rusch NJ (2006) Vascular calcium channels and high blood pressure: pathophysiology and therapeutic implications. Vasc Pharmacol 44(3):131–142. https://doi.org/10.1016/j.vph.2005.10.005, PMID 16427812

Sturrock A, Cahill B, Norman K, Huecksteadt TP, Hill K, Sanders K et al (2006) Transforming growth factor-β1 induces Nox4 NAD(P)H oxidase and reactive oxygen species-dependent proliferation in human pulmonary artery smooth muscle cells. Am J Physiol Lung Cell Mol Physiol 290(4):L661–L673. https://doi.org/10.1152/ajplung.00269.2005, PMID 16227320

Sui H, Zhao J, Zhou L, Wen H, Deng W, Li C et al (2017) Tanshinone IIA inhibits β-catenin/VEGF-mediated angiogenesis by targeting TGF-β1 in normoxic and HIF-1α in hypoxic microenvironments in human colorectal cancer. Cancer Lett 403:86–97. https://doi.org/10.1016/j.canlet.2017.05.013, PMID 28602978

Tabas I, Bornfeldt KE (2016) Macrophage phenotype and function in different stages of atherosclerosis. Circ Res 118(4):653–667. https://doi.org/10.1161/CIRCRESAHA.115.306256, PMID 26892964

Tang F, Wu X, Wang T, Wang P, Li R, Zhang H et al (2007) Tanshinone II A attenuates atherosclerotic calcification in rat model by inhibition of oxidative stress. Vasc Pharmacol 46(6):427–438. https://doi.org/10.1016/j.vph.2007.01.001, PMID 17337361

Wang JW, Wu JY (2010) Tanshinone biosynthesis in Salvia miltiorrhiza and production in plant tissue cultures. Appl Microbiol Biotechnol 88(2):437–449. https://doi.org/10.1007/s00253-010-2797-7, PMID 20694462

Wang X, Morris-Natschke SL, Lee KH (2007) New developments in the chemistry and biology of the bioactive constituents of Tanshen. Med Res Rev 27(1):133–148. https://doi.org/10.1002/med.20077, PMID 16888751

Wang J, Jiang Q, Wan L, Yang K, Zhang Y, Chen Y et al (2013) Sodium tanshinone IIA sulfonate inhibits canonical transient receptor potential expression in pulmonary arterial smooth muscle from pulmonary hypertensive rats. Am J Respir Cell Mol Biol 48(1):125–134. https://doi.org/10.1165/rcmb.2012-0071OC, PMID 23065131

Wang et al (2015) Herba epimedii: an ancient chinese herbal medicine in the prevention and treatment of osteoporosis. Curr Pharm Des 22(3):328–349. https://doi.org/10.2174/1381612822666151112145907

Wang CH, Zheng LP, Tian H, Wang JW (2016a) Synergistic effects of ultraviolet-B and methyl jasmonate on tanshinone biosynthesis in Salvia miltiorrhiza hairy roots. J Photochem Photobiol B 159:93–100. https://doi.org/10.1016/J.JPHOTOBIOL.2016.01.012, PMID 27043259

Wang L, Li Y, Guo Y, Ma R, Fu M, Niu J et al (2016b) Herba Epimedii: an ancient Chinese herbal medicine in the prevention and treatment of osteoporosis. Curr Pharm Des 22(3):328–349. https://doi.org/10.2174/1381612822666151112145907, PMID 26561074

Wang N, Chang Y, Chen L, Guo YJ, Zhao YS, Guo QH et al (2017) Tanshinone IIA protects against chronic intermittent hypoxia-induced myocardial injury via activating the endothelin 1 pathway. Biomed Pharmacother 95:1013–1020. https://doi.org/10.1016/j.biopha.2017.08.036, PMID 28922718

Wang L, Cheng L, Zhang B, Wang N, Wang F (2019) Tanshinone prevents alveolar bone loss in ovariectomized osteoporosis rats by up-regulating phosphoglycerate dehydrogenase. Toxicol Appl Pharmacol 376:9–16. https://doi.org/10.1016/j.taap.2019.05.014, PMID 31108107

Wang X, Yang Y, Liu X, Gao X (2020) Pharmacological properties of tanshinones, the natural products from Salvia miltiorrhiza. Adv Pharmacol 87:43–70. https://doi.org/10.1016/bs.apha.2019.10.001

Wu DM, Wang YJ, Han XR, Wen X, Li L, Xu L et al (2018) Retracted. J Cell Mol Med 22(6): 3058–3072. https://doi.org/10.1111/jcmm.13557, PMID 29524303

Xu W, Yang J, Wu LM (2009) Cardioprotective effects of tanshinone IIA on myocardial ischemia injury in rats. Pharmazie 64(5):332–336, PMID 19530445

Xu Z, Chen L, Xiao Z, Zhu Y, Jiang H, Jin Y et al (2018) Potentiation of the anticancer effect of doxorubicinin drug-resistant gastric cancer cells by tanshinone IIA. Phytomedicine 51:58–67. https://doi.org/10.1016/j.phymed.2018.05.012, PMID 30466628

Yadav S, Kalra N, Ganju L, Singh M (2017) Activator protein-1 (AP-1): a bridge between life and death in lung epithelial (A549) cells under hypoxia. Mol Cell Biochem 436(1–2):99–110. https://doi.org/10.1007/s11010-017-3082-1, PMID 28589371

Yadav S, Singh M, Singh SN, Kumar B (2020) Tanshinone IIA pretreatment promotes cell survival in human lung epithelial cells under hypoxia via AP-1-Nrf2 transcription factor. Cell Stress Chaperones 25(3):427–440. https://doi.org/10.1007/s12192-020-01083-3, PMID 32144684

Yang D, Ma P, Liang X, Wei Z, Liang Z, Liu Y et al (2012) PEG and ABA trigger methyl jasmonate accumulation to induce the MEP pathway and increase tanshinone production in Salvia miltiorrhiza hairy roots. Physiol Plant 146(2):173–183. https://doi.org/10.1111/j.1399-3054.2012.01603.x, PMID 22356467

Yang JX, Pan YY, Ge JH, Chen B, Mao W, Qiu YG et al (2016) Tanshinone II A attenuates TNF-α-induced expression of VCAM-1 and ICAM-1 in endothelial progenitor cells by blocking activation of NF-κB. Cell Physiol Biochem 40(1–2):195–206. https://doi.org/10.1159/000452537, PMID 27855363

Yintao Y (2010) Effects of cryptotanshinone on proliferation and apoptosis of Hela cell line of cervical cancer. Chin J Chin Matera Med 35:35. https://doi.org/10.4268/cjcmm20100125

Zhang Y, Zhang L, Chu W, Wang B, Zhang J, Zhao M et al (2010) Tanshinone IIA inhibits miR-1 expression through p38 MAPK signal pathway in post-infarction rat cardiomyocytes. Cell Physiol Biochem 26(6):991–998. https://doi.org/10.1159/000324012, PMID 21220930

Zhang J, Wang J, Jiang JY, Liu SD, Fu K, Liu HY (2014) Tanshinone IIA induces cytochrome c-mediated caspase cascade apoptosis in A549 human lung cancer cells via the JNK pathway. Int J Oncol 45(2):683–690. https://doi.org/10.3892/ijo.2014.2471, PMID 24888720

Zhang N, Dong M, Luo Y, Zhao F, Li Y (2018a) Danshensu prevents hypoxic pulmonary hypertension in rats by inhibiting the proliferation of pulmonary artery smooth muscle cells via TGF-β-smad3-associated pathway. Eur J Pharmacol 820:1–7. https://doi.org/10.1016/j.ejphar.2017.12.010, PMID 29221952

Zhang Y, Zhang S, Chen X (2018b) Tanshinone IIA protects against cardiac fibrosis through inhibition of β-tubulin expression. J Biol Regul Homeost Agents 32(6):1451–1455, PMID 30574749

Zhou L, Zuo Z, Chow MSS (2005) Danshen: an overview of its chemistry, pharmacology, pharmacokinetics, and clinical use. J Clin Pharmacol 45(12):1345–1359. https://doi.org/10.1177/0091270005282630, PMID 16291709

Zhou D, Li Z, Zhang L, Zhan C (2012) Inhibitory effect of tanshinone II A on TGF II-β1-induced cardiac fibrosis. J Huazhong Univ Sci Technolog Med Sci 32(6):829–833. https://doi.org/10.1007/s11596-012-1042-2, PMID 23271281